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Explorer

Open Editors

DefaultAccessMain.java

ProtectedAccessMain.java

SingleInheritanceMain.java

HierarchicalMain.java

MultilevelMain.java

MedicalStaffMain.java

InheritanceConceptMain.java

ShapeMain.java

HospitalStaffMain.java

Lab Sheet 05

DefaultAccessMain.java

Doctor.class

Employee.class

HierarchicalMain.java

HospitalStaffMain.class

HospitalStaffMain.java

InheritanceConceptMain.java

MedicalStaffMain.java

MultilevelMain.java

Person.class

Pharmacist.class

PharmacyEmployeeMain.java

ProtectedAccessMain.java

ShapeMain.java

SingleInheritanceMain.java

StaffManager.class

Outline

Timeline

Servers

Java Projects

Lab Sheet 05

DefaultAccessMain.java

ProtectedAccessMain.java

SingleInheritanceMain.java

HierarchicalMain.java

Run | Debug

```
1 public class DefaultAccessMain {
2     public static void main(String[] args) {
3
4         Pharmacy pharmacy = new Pharmacy();
5
6         // Default/package-private members can be accessed
7         // because both classes are in the same package.
8         pharmacy.medicineName = "Paracetamol";
9         pharmacy.price = 25.50;
10
11         pharmacy.displayMedicine();
12     }
13 }
14
15 // Non-public Pharmacy class
16 class Pharmacy {
17
18     // Default/package-private variables
19     String medicineName;
20     double price;
21
22     // Default/package-private constructor
23     Pharmacy() {
```

Problems

Output

Debug Console

Terminal

Ports

```
PS C:\Users\HP-PC\OneDrive\Desktop\Lab Sheet 05> & 'C:\Program Files\Java\jdk-26.0.2.1\bin\java.exe'
'--enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\HP-PC\AppData\Roaming\C
ode\User\workspaceStorage\3f00a0488576b5efad804e59c1a6fdaa\redhat.java\jdt_ws\Lab Sheet 05_7a66c801\b
in' 'DefaultAccessMain'
Medicine Name: Paracetamol
Price: 25.5
PS C:\Users\HP-PC\OneDrive\Desktop\Lab Sheet 05>
```

Run... 🛑

Run: Protec...

📄 powershell

Run: Hierar...

📄 powershell

Run: Inherit...

Run: Shape...

📄 powershell

Chat

+ 🔧 ⋮ 🗑️

Build with Agent

AI responses may be inaccurate

[Generate Agent instructions](#) to onboard AI onto your codebase.

+ 📄 DefaultAccessMain.java

Describe what to build

+ 🧑 Agent Models ⚖️ 🗣️ ⬆️

📄 Local 🛡️ Default permissions

Ln 33, Col 2

Spaces: 4

UTF-8

CRLF

{ } Java

🔄 Go Live

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Open Editors

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Lab Sheet 05

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StaffManager.class

Outline

Timeline

Servers

Java Projects

Lab Sheet 05

ProtectedAccessMain.java

Pharmacist

```
12 class HospitalStaff {
18     protected HospitalStaff(int staffId, String staffName) {
21     }
22
23     // Protected method
24     protected void displayStaff() {
25         System.out.println("Staff ID: " + staffId);
26         System.out.println("Staff Name: " + staffName);
27     }
28 }
29
30 // Child class
31 class Pharmacist extends HospitalStaff {
32
33     private String licenseNo;
34
35     Pharmacist(int staffId, String staffName, String licenseNo) {
36
37         // Calling parent constructor
38         super(staffId, staffName);
39
40         this.licenseNo = licenseNo;
41     }
42 }
```

Problems

Output

Debug Console

Terminal

Ports

```
PS C:\Users\HP-PC\OneDrive\Desktop\Lab Sheet 05> & 'C:\Program Files\Java\jdk-26.0.2.1\bin\java.exe'
'--enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\HP-PC\AppData\Roaming\C
ode\User\workspaceStorage\3f00a0488576b5efad804e59c1a6fdaa\redhat.java\jdt_ws\Lab Sheet 05_7a66c801\b
in' 'ProtectedAccessMain'
Staff ID: 101
Staff Name: Bhavna
License No: LTC12345
PS C:\Users\HP-PC\OneDrive\Desktop\Lab Sheet 05> 
```

Run: Default...

Run: ...

powershell

Run: Hierar...

powershell

Run: Inherit...

Run: Shape...

powershell

Chat

Build with Agent

AI responses may be inaccurate

Generate Agent Instructions to onboard AI onto your codebase.

ProtectedAccessMain.java

Describe what to build

Agent Models

Local Default permissions

Ln 50, Col 2 Spaces: 4 UTF-8 CRLF {} Java Go Live

Explorer

Open Editors

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Lab Sheet 05

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PharmacyEmployeeMain.java

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ShapeMain.java

SingleInheritanceMain.java

StaffManager.class

Outline

Timeline

Servers

Java Projects

Lab Sheet 05

SingleInheritanceMain.java > Employee > employeeName

```
16 class Employee {
21
22     Employee(int employeeId, String employeeName, double salary) {
23         this.employeeId = employeeId;
24         this.employeeName = employeeName;
25         this.salary = salary;
26     }
27
28     void displayEmployee() {
29         System.out.println("Employee ID: " + employeeId);
30         System.out.println("Employee Name: " + employeeName);
31         System.out.println("Salary: " + salary);
32     }
33 }
34
35 // Child class
36 class Pharmacist extends Employee {
37
38     private String licenseNo;
39
40     Pharmacist(int employeeId, String employeeName,
41               double salary, String licenseNo) {
42
43         // Initialize parent class
```

Problems

Output

Debug Console

Terminal

Ports

PS C:\Users\HP-PC\OneDrive\Desktop\Lab Sheet 05> java SingleInheritanceMain.java

Employee ID: 101

Employee Name: Bhavna

Salary: 45000.0

License No: LTC12345

PS C:\Users\HP-PC\OneDrive\Desktop\Lab Sheet 05>

Run: Default...

Run: Protec...

pow... [] []

Run: Hierar...

powershell

powershell

Run: Inherit...

Run: Shape...

powershell

Chat

Build with Agent

AI responses may be inaccurate

Generate Agent instructions to onboard AI onto your codebase.

SingleInheritanceMain.java

Describe what to build

+ Agent Models

Local Default permissions

Ln 19, Col 35 Spaces: 4 UTF-8 CRLF { } Java Go Live

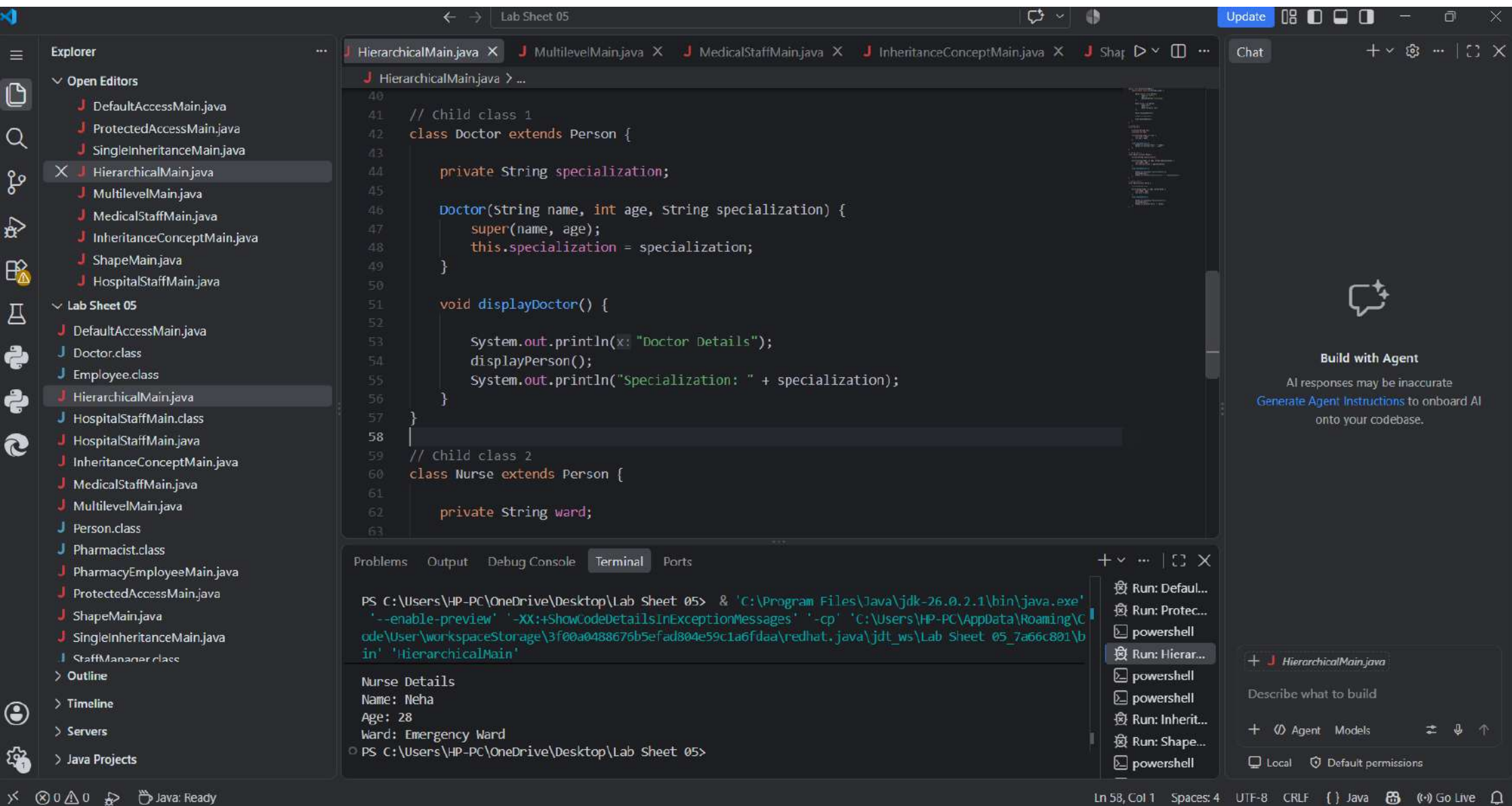
The screenshot displays an IDE interface with the following components:

- Explorer Panel (Left):** Lists project files. Under "Lab Sheet 05", the file `SingleInheritanceMain.java` is selected.
- Editor Panel (Center):** Shows the code for `SingleInheritanceMain.java`. The `Pharmacist` class is defined, extending `Employee`. It includes a `licenseNo` attribute and a `displayPharmacist()` method that calls `displayEmployee()` and prints the license number.

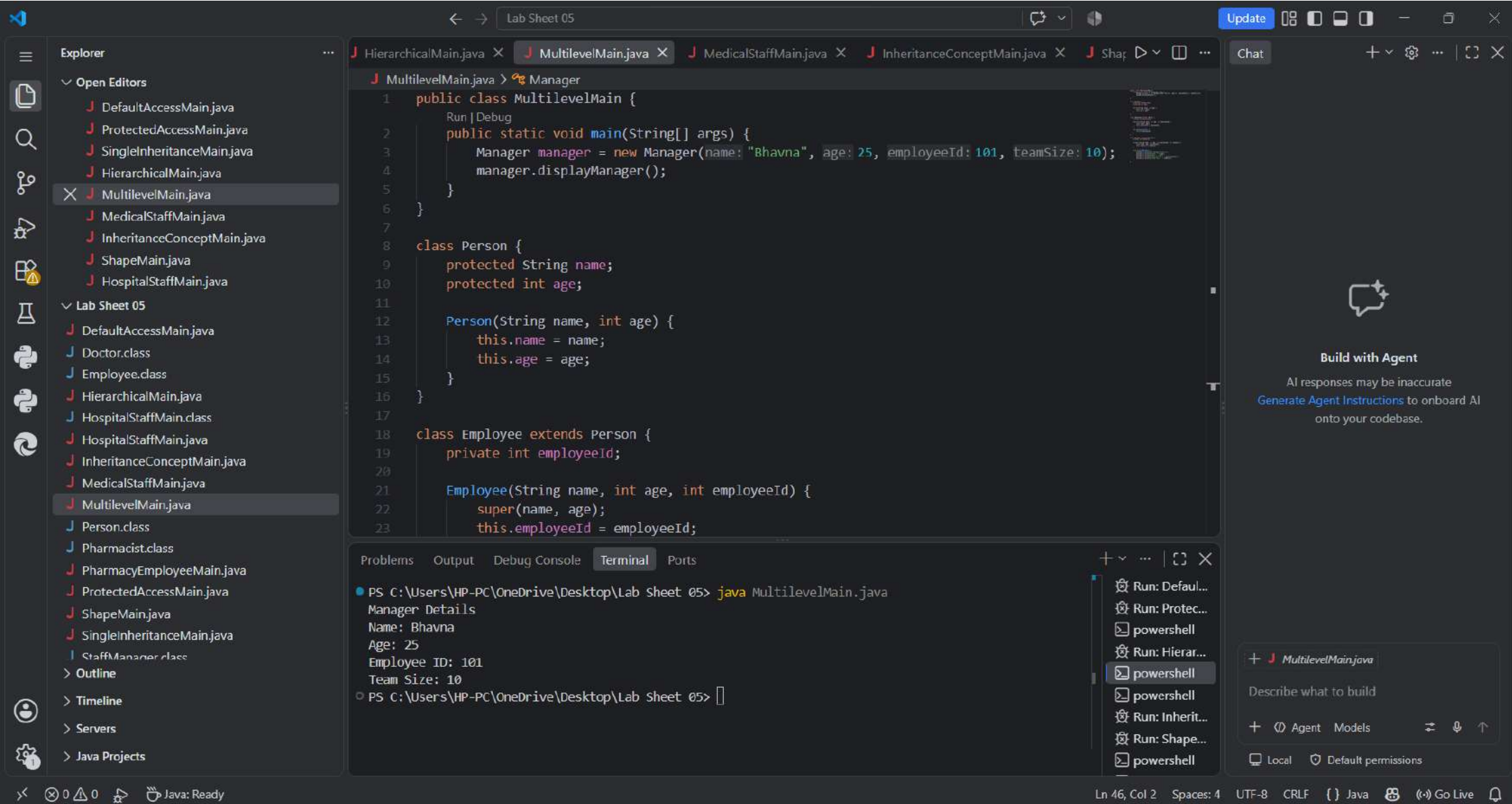
```
36 class Pharmacist extends Employee {
37
38     private String licenseNo;
39
40     Pharmacist(int employeeId, String employeeName,
41               double salary, String licenseNo) {
42
43         // Initialize parent class
44         super(employeeId, employeeName, salary);
45
46         // Initialize child class variable
47         this.licenseNo = licenseNo;
48     }
49
50     void displayPharmacist() {
51
52         displayEmployee();
53
54         System.out.println("License No: " + licenseNo);
55     }
56 }
```
- Terminal Panel (Bottom):** Shows the execution of `SingleInheritanceMain.java`. The output is:

```
PS C:\Users\HP-PC\OneDrive\Desktop\Lab Sheet 05> java SingleInheritanceMain.java
Employee ID: 101
Employee Name: Bhavna
Salary: 45000.0
License No: LIC12345
PS C:\Users\HP-PC\OneDrive\Desktop\Lab Sheet 05>
```
- Right Panel:** Contains a "Chat" section with a "Build with Agent" button and a prompt to "Describe what to build".

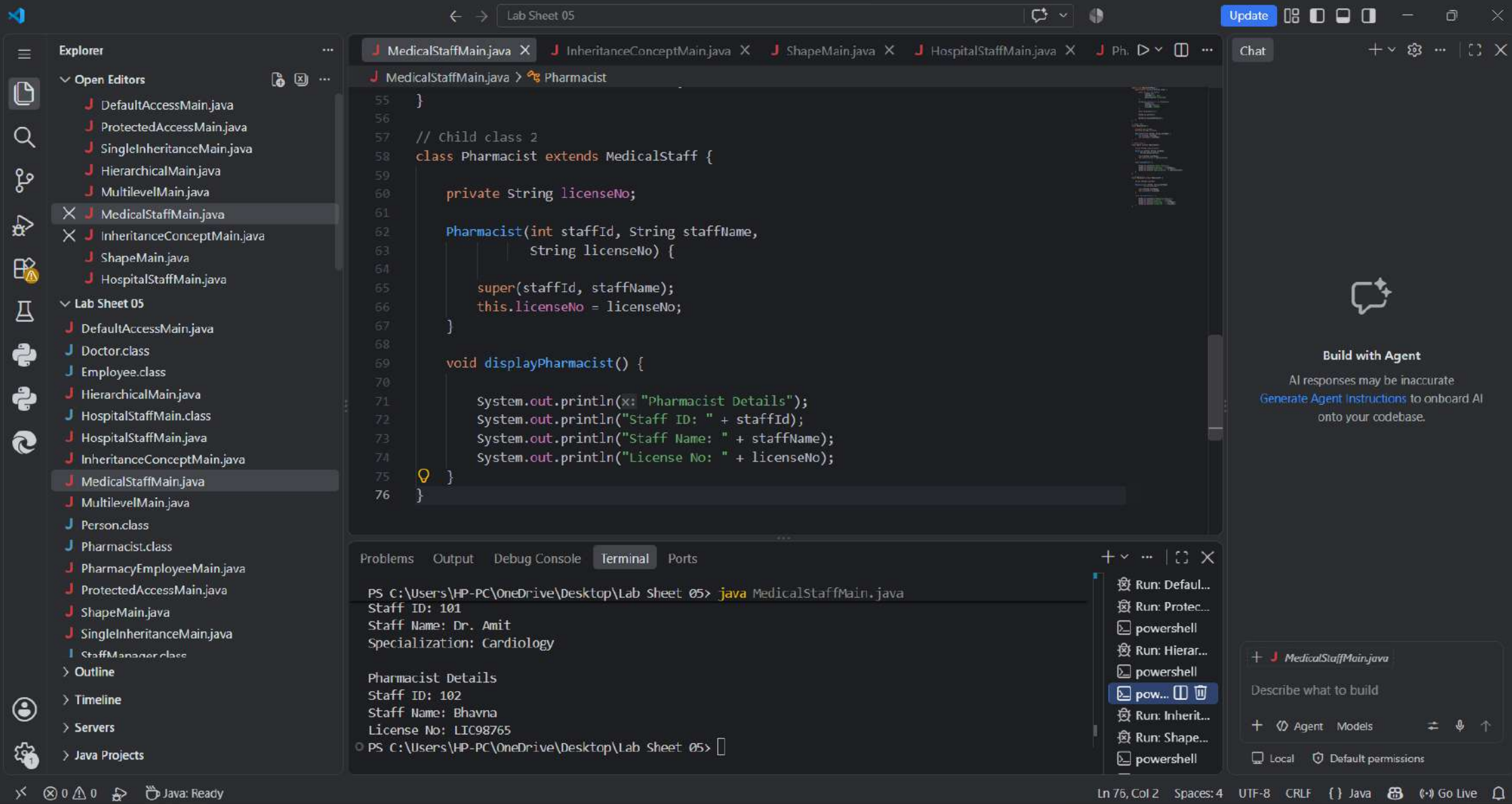
The screenshot displays the IntelliJ IDEA IDE interface. On the left, the 'Explorer' panel shows a project structure with various Java files. The 'Open Editors' section lists several files, with 'HierarchicalMain.java' selected. The 'Lab Sheet 05' section also lists files, including 'HierarchicalMain.java'. The main editor window shows the code for 'HierarchicalMain.java', which defines a 'Person' class and a 'Nurse' class extending it. The 'Nurse' class has a 'displayPerson()' method that prints the name and age. The 'Terminal' panel at the bottom shows the command to run the application, and the output displays 'Nurse Details' with Name: Neha, Age: 28, and Ward: Emergency Ward. The right sidebar shows the 'Build with Agent' section with a chat interface.

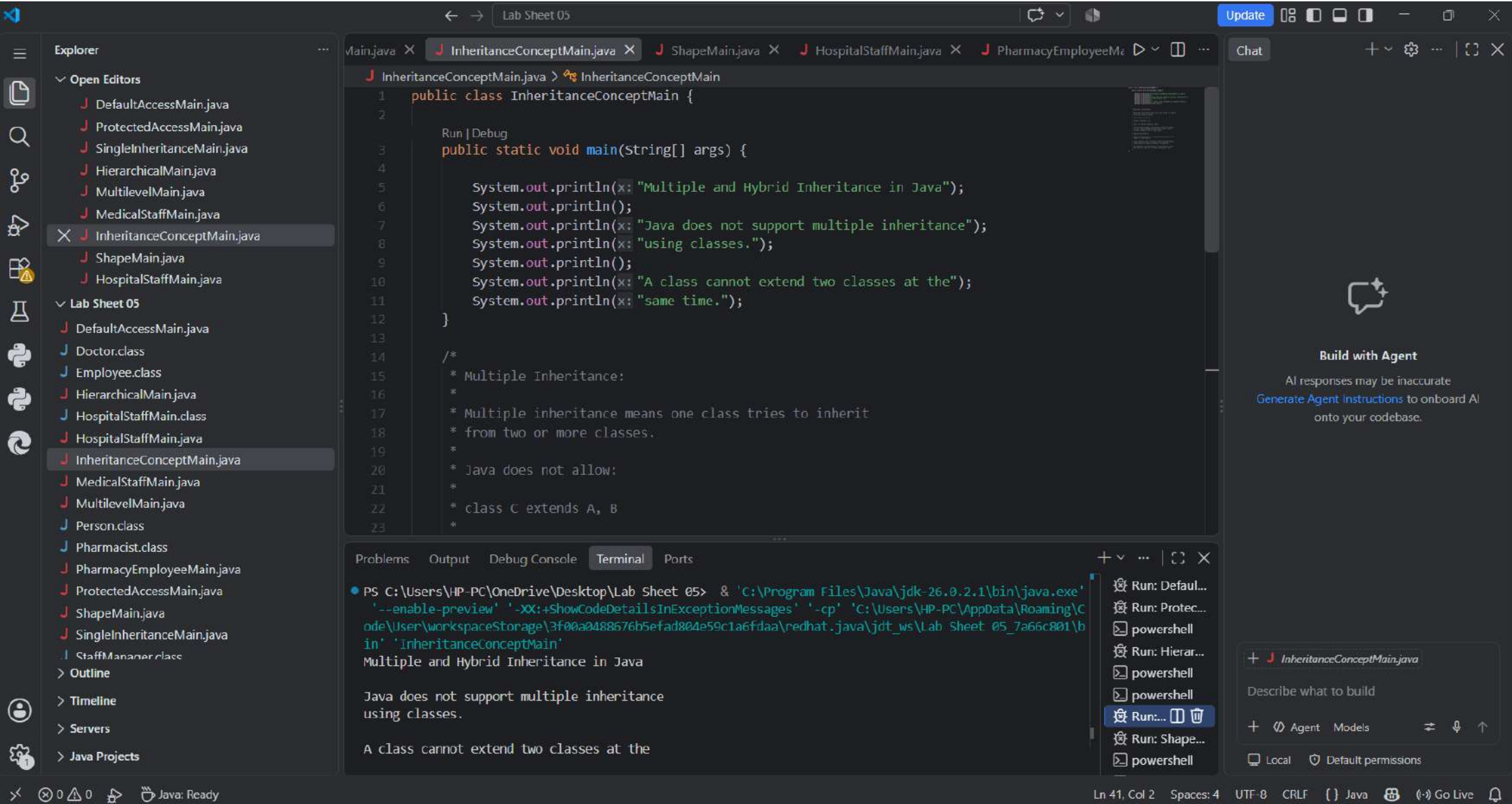


The image shows a screenshot of an IDE (likely IntelliJ IDEA) with a dark theme. The interface is divided into several panels. On the left is the 'Explorer' panel showing a project structure with files like 'DefaultAccessMain.java', 'ProtectedAccessMain.java', 'SingleInheritanceMain.java', 'HierarchicalMain.java' (selected), 'MultilevelMain.java', 'MedicalStaffMain.java', 'InheritanceConceptMain.java', 'ShapeMain.java', and 'HospitalStaffMain.java'. Below this is the 'Lab Sheet 05' section with files like 'DefaultAccessMain.java', 'Doctor.class', 'Employee.class', 'HierarchicalMain.java' (selected), 'HospitalStaffMain.class', 'HospitalStaffMain.java', 'InheritanceConceptMain.java', 'MedicalStaffMain.java', 'MultilevelMain.java', 'Person.class', 'Pharmacist.class', 'PharmacyEmployeeMain.java', 'ProtectedAccessMain.java', 'ShapeMain.java', 'SingleInheritanceMain.java', 'StaffManager.class', 'Outline', 'Timeline', 'Servers', and 'Java Projects'. The main editor window displays the code for 'HierarchicalMain.java', showing a Java class hierarchy where 'Nurse' extends 'Person'. The code includes a constructor 'Nurse(String name, int age, String ward)' and a method 'displayNurse()' that prints 'Nurse Details' and 'Ward: ' + ward. Below the editor is a 'Terminal' panel showing the command 'PS C:\Users\HP-PC\OneDrive\Desktop\Lab Sheet 05> java -cp .\bin\java.exe -enable-preview -XX:+ShowCodeDetailsInExceptionMessages -cp 'C:\Users\HP-PC\AppData\Roaming\Code\User\workspaceStorage\3f00a0488676b5efad804e59c1a6fdaa\redhat.java\jdt_ws\Lab Sheet 05_7a66c801\bin' HierarchicalMain' and the output 'Nurse Details', 'Name: Neha', 'Age: 28', 'Ward: Emergency Ward'. On the right side, there is a 'Chat' panel with a 'Build with Agent' section, which includes a warning 'AI responses may be inaccurate' and a link to 'Generate Agent Instructions to onboard AI onto your codebase.' Below this is a search bar for 'HierarchicalMain.java' and a section for 'Describe what to build' with a '+ Agent Models' button and a 'Local' button.



The image shows a screenshot of the Visual Studio Code IDE. On the left, the Explorer sidebar displays a project structure with several Java files. The 'MedicalStaffMain.java' file is selected and open in the main editor. The editor shows the code for a 'Pharmacist' class that extends 'MedicalStaff'. The code includes a 'displayDoctor()' method in the parent class and a 'displayPharmacist()' method in the child class. The Terminal at the bottom shows the execution of 'java MedicalStaffMain.java', resulting in two sets of output: one for a Doctor (Staff ID: 101, Name: Dr. Amit, Specialization: Cardiology) and one for a Pharmacist (Staff ID: 102, Name: Bhavna, License No: LIC98765). The Chat sidebar on the right features a 'Build with Agent' section, suggesting AI-assisted development.





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Explorer

▼ Open Editors

DefaultAccessMain.java

ProtectedAccessMain.java

SingleInheritanceMain.java

HierarchicalMain.java

MultilevelMain.java

MedicalStaffMain.java

✕ InheritanceConceptMain.java

ShapeMain.java

HospitalStaffMain.java

▼ Lab Sheet 05

DefaultAccessMain.java

Doctor.class

Employee.class

HierarchicalMain.java

HospitalStaffMain.class

HospitalStaffMain.java

InheritanceConceptMain.java

MedicalStaffMain.java

MultilevelMain.java

Person.class

Pharmacist.class

PharmacyEmployeeMain.java

ProtectedAccessMain.java

ShapeMain.java

SingleInheritanceMain.java

StaffManager.class

> Outline

> Timeline

> Servers

> Java Projects

Lab Sheet 05

main.java ✕ InheritanceConceptMain.java ✕ ShapeMain.java ✕ HospitalStaffMain.java ✕ PharmacyEmployeeMain.java

InheritanceConceptMain.java > InheritanceConceptMain

1 public class InheritanceConceptMain {
23 *
24 * This is invalid syntax in Java.
25 *
26 * Java avoids multiple inheritance of classes because
27 * it can create ambiguity when both parent classes
28 * contain methods with the same name.
29 *
30 * Hybrid Inheritance:
31 *
32 * Hybrid inheritance is a combination of two or more
33 * types of inheritance.
34 *
35 * Java commonly uses interfaces when multiple/hybrid
36 * inheritance of type or behavior is required.
37 *
38 * No interface is implemented in this program because
39 * the question asks for concept explanation only.
40 */
41 }

Problems

Output

Debug Console

Terminal

Ports

PS C:\Users\HP-PC\OneDrive\Desktop\Lab Sheet 05> & 'C:\Program Files\Java\jdk-26.0.2.1\bin\java.exe' '-enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\HP-PC\AppData\Roaming\Code\User\workspaceStorage\3f00a0488676b5efad804e59c1a6fdaa\redhat.java\jdt_ws\Lab Sheet 05_7a66c801\bin' 'InheritanceConceptMain'
Java does not support multiple inheritance using classes.

A class cannot extend two classes at the same time.
PS C:\Users\HP-PC\OneDrive\Desktop\Lab Sheet 05>

Run: Default...

Run: Protec...

powershell

Run: Hierar...

powershell

powershell

Run:...

Run: Shape...

powershell

+ InheritanceConceptMain.java

Describe what to build

+ Agent Models

Local Default permissions

Ln 41, Col 2 Spaces: 4 UTF-8 CRLF {} Java Go Live